

MORRIS, MN, USA

WMO: 726565

Lat: 45.566N

Lon: 95.968W

Elev: 344

StdP: 97.26

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.4	(c) -24.1	(d) -30.8	(e) 0.2	(f) -27.3	(g) -27.6	(h) 0.3	(i) -23.9	(j) 15.8	(k) -11.7	(l) 14.5	(m) -12.4	(n) 3.8	(o) 320	(p) 0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.1	23.2	30.0	22.2	28.0	21.1	25.3	29.6	24.0	28.0	23.0	26.7	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.6	27.8	22.7	18.1	26.5	22.1	17.4	26.0	79.3	29.2	74.2	28.0	69.9	27.0	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	11.9	10.7	-29.9	34.5	3.5	1.7	-32.5	35.6	-34.5	36.6	-36.5	37.6	-39.1	38.8
			-30.2	27.1	3.4	1.4	-32.6	28.1	-34.6	28.9	-36.5	29.6	-39.0	30.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-11.4	-9.5	-1.8	6.4	13.5	19.3	21.5	19.9	15.9	8.2	-0.4	-8.2
	DBStd	12.91	7.75	7.44	7.16	5.63	4.93	3.76	3.23	3.12	4.59	5.19	6.32	6.82
	HDD10.0	2714	662	545	369	133	22	1	0	0	6	98	316	563
	HDD18.3	4709	921	778	624	358	163	32	9	20	101	318	562	822
	CDD10.0	1328	0	0	3	26	131	281	357	306	182	41	3	0
	CDD18.3	282	0	0	0	1	14	62	108	68	27	2	0	0
	CDH23.3	2731	0	0	1	21	202	594	1009	587	279	37	0	0
	CDH26.7	808	0	0	0	6	66	183	323	147	74	9	0	0
(14) Wind	WSAvg	4.7	5.2	5.2	5.3	5.5	5.4	4.5	3.6	3.5	4.1	4.9	5.0	5.0
(15) Precipitation	PrecAvg	649	20	17	34	60	73	102	99	81	62	55	29	17
	PrecMax	866	68	42	96	157	171	194	231	189	152	234	93	88
	PrecMin	251	0	2	4	4	7	12	27	5	6	2	2	0
	PrecStd	142	15	11	20	35	39	48	49	44	34	49	23	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	9.8	18.9	26.3	32.3	32.9	33.7	32.4	32.2	27.7	18.9	8.8
		MCWB	3.1	7.9	12.5	15.3	19.2	22.5	25.3	25.0	22.4	17.1	12.8	4.9
	2%	DB	2.6	5.2	14.2	22.3	28.0	31.1	32.0	29.8	28.7	22.8	14.8	4.0
		MCWB	1.3	3.3	10.3	13.1	17.8	21.8	24.7	23.6	20.5	14.8	9.8	2.0
	5%	DB	1.2	2.6	11.0	18.8	25.3	28.7	29.9	27.9	27.1	19.0	11.2	2.4
		MCWB	0.4	1.6	8.1	11.5	16.4	20.5	23.2	21.9	19.6	13.3	8.0	1.3
	10%	DB	-0.9	1.2	7.5	16.2	22.5	27.1	28.0	27.1	24.0	17.0	7.8	1.0
		MCWB	-1.3	0.5	5.6	10.1	14.9	19.4	22.1	21.3	18.0	11.8	5.4	0.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	7.5	14.1	16.8	21.7	25.1	27.4	26.4	24.0	19.2	14.2	5.4
		MCDB	5.7	8.7	17.2	24.2	28.1	30.4	30.7	30.1	28.6	24.6	18.0	8.0
	2%	WB	1.5	3.6	11.1	14.4	19.6	23.6	25.8	24.6	22.6	16.4	10.4	2.6
		MCDB	2.6	5.0	14.2	19.9	24.9	28.5	30.0	28.4	27.0	21.2	14.0	4.0
	5%	WB	0.3	1.8	7.9	12.3	17.8	22.2	24.6	23.4	21.1	14.1	7.9	1.3
		MCDB	1.1	2.5	10.0	17.5	22.8	26.7	28.6	26.7	24.4	18.1	10.8	2.2
	10%	WB	-1.2	0.3	5.4	10.5	16.1	20.7	23.4	22.2	19.4	12.3	5.5	0.0
		MCDB	-0.8	1.1	7.7	15.2	21.2	24.9	27.0	25.6	23.2	15.8	8.0	0.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	9.6	11.5	12.1	11.4	11.5	11.8	12.7	11.0	9.6	8.7
		MCDBR	10.3	10.5	14.3	17.3	16.3	14.1	13.1	13.2	15.4	16.1	14.5	11.2
		MCWBR	9.1	9.0	9.9	9.7	8.3	6.6	6.8	7.3	7.9	9.7	9.7	9.1
	5% WB	MCDBR	9.7	9.6	13.4	15.3	13.5	11.6	11.7	11.7	12.4	14.0	13.6	10.3
		MCWBR	8.6	8.3	9.6	9.4	8.0	6.7	7.2	7.5	7.6	9.1	9.5	8.6
(40) Clear-Sky Solar Irradiance	taub		0.268	0.282	0.308	0.358	0.381	0.387	0.388	0.393	0.359	0.312	0.281	0.261
	taud		2.415	2.386	2.431	2.344	2.343	2.378	2.363	2.335	2.418	2.533	2.530	2.454
	Ebn at Noon		852	906	927	901	887	880	874	855	858	863	837	826
	Edh at Noon		70	88	98	117	121	118	118	117	99	76	63	61
(44) All-Sky Solar Radiation	RadAvg		1.73	2.82	3.81	4.78	5.46	6.10	6.48	5.42	4.15	2.62	1.69	1.23
	RadStd		0.20	0.31	0.34	0.38	0.40	0.38	0.24	0.45	0.30	0.36	0.17	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page